Chapter 21

Health Technology Assessment—Policy Objectives and Principles of System Design

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Learning Objectives

Objective 21.1 Overview of health technology assessment (HTA) from a policy perspective.
Objective 21.2 Describe different paradigms in the assessment of health technologies.
Objective 21.3 Describe a high-level HTA process.
Objective 21.4 Explain HTA system design in current practice.

OBJECTIVE 21.1. OVERVIEW OF HEALTH TECHNOLOGY ASSESSMENT FROM A POLICY PERSPECTIVE

Meaning and Purpose of Health Technology Assessment

Health technology assessment (HTA) is commonly defined as a multidisciplinary approach for the systematic assessment of medical technologies regarding their effectiveness, appropriateness, efficiency, and social and ethical aspects and implications.[1] It is usually conducted by an interdisciplinary group using analytical frameworks, drawing from a variety of methods.[2] By systematically assessing health technologies, HTA fulfills an important decision support role with regard to decisions on a rational allocation of healthcare resources. Thus, it can support health policy in general and, very commonly, pricing and funding decisions in particular. It is supposed to provide responsible bodies and organizations with timely, accurate, and sound information on newly introduced medical technologies. Besides this, HTA can also support decisions on already reimbursed health technologies, for example, pricing and funding reviews.[3]

In principle, HTA can evaluate all medical technologies, including pharmaceuticals, diagnostics, medical devices, surgical procedures, or even public health interventions, such as disease screening or vaccination programs. In most countries, however, HTA is focused mainly on pharmaceuticals (most commonly new active substances). There are three main reasons for this:

- pharmaceuticals are developed, registered, and funded on the basis of evidence generated in clinical trials and, increasingly, real-world evidence;
- they are highly standardized technologies with well-defined intellectual property rights (e.g., patents); and
- they are often associated with high costs or significant impact on public health budgets, which makes payers (budget holders) conscious about the financial implications of pricing or funding decisions.

As discussed below, HTA has often been interpreted as a form of economic analysis focused on cost-effectiveness and budget impact. *Fourth hurdle*,[4] a previously very popular expression, also refers to this suggesting that HTA in general and cost-effectiveness analysis in particular constitute a fourth dimension of critical assessment besides the three main aspects covered in regulatory approval processes: efficacy, safety, and quality. HTA is, however, not limited to, economic analysis.

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Clinical assessment is a core component of most HTA systems and many HTA systems consider other aspects in line with the broader academic definition of HTA (e.g., ethical and social considerations).

Traditionally regulatory approval processes and HTA systems have not been interlinked and they have focused on different aspects or comparisons. For instance, a clinical trial design which was approved by regulatory bodies may have been judged as inappropriate or insufficient during a subsequent HTA assessment. An example is the acceptance of placebo-controlled studies for regulatory purposes as opposed to HTA bodies strongly favoring active comparator studies. Similarly, regulatory agencies may grant marketing authorization to a product based on Phase II data which may not be sufficient for HTA bodies to issue positive recommendations. The regulatory and HTA bodies have recognized the necessity of interacting more closely with each other and with technology owners (pharmaceutical companies) by providing harmonized guidance (e.g., early advice) during clinical development.[5,6]

Policy Objectives Related to Health Technology Assessment

Why do healthcare policy-makers or payers (budget holders) rely on HTA? The economist's answer lies in two interrelated concepts in health economics: *supply-induced demand* and *opportunity costs*.[7] A political scientist or a sociologist will add other factors, too.

In simple terms, *supply-induced demand* means that medical professionals and patients will want to use any new health technology that is launched on the healthcare market. The main reason is that new technologies offer, or at least claim, better treatment possibilities and results. In the case of technologies funded by public or private (third-party) payers for patients, low or zero price sensitivity also reinforces supply-induced demand, which means that payers would need ample or even unlimited resources to pay for all healthcare innovations. This is certainly unlikely to ever happen—healthcare budgets are limited even in affluent countries.

Thus, by paying for one health technology, policy-makers and payers will have to forgo the purchase of, or provision of funding to, other health technologies. This situation, whereby the selection of one opportunity rules out the selection of another opportunity for the lack of resources, is called "*opportunity cost*" in economics. Policy-makers and payers execute their role and mandate properly if they minimize opportunity costs in their purchasing decisions by identifying and paying for technologies which have the highest *value* compared with other available and already funded technologies. For this, policy-makers and payers will rely on structured evaluation methods to assess the value of new health technologies in comparison to already available and funded health technologies. This is the economist's explanation for the development and institutionalization of HTA across healthcare systems (Fig. 21.1).

A political scientist will add that structured evaluation methods are likely to increase the transparency and accountability of healthcare funding decisions, which is a significant guarantee for higher-quality health services.[8] From this perspective, the use of HTA decreases the political vulnerability of individual decision-makers through reinforcing institutional decision responsibilities over individual responsibilities. Political science, however, also points out that HTA can also be used or misused for cost containment purposes and/or for limiting patients' access to new health technologies.

OBJECTIVE 21.2. DESCRIBE DIFFERENT PARADIGMS IN THE ASSESSMENT OF HEALTH TECHNOLOGIES

A crucial issue in HTA is that *value* is a concept with several possible definitions. HTA approaches differ significantly in the way they consider and assess the value. These differences have given rise to diverging HTA systems and methodologies around the world. In HTA terminology, these are sometimes referred to as *value frameworks*.[9]

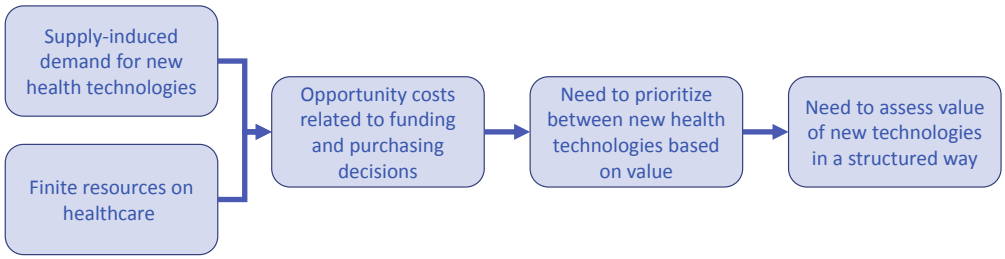


FIGURE 21.1 An economist's explanation of the need for health technology assessment.

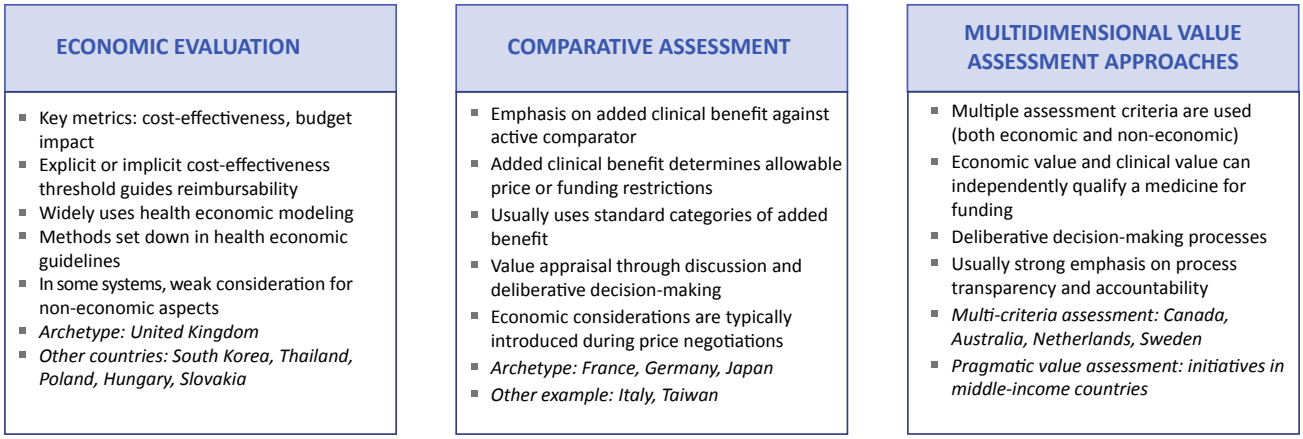


FIGURE 21.2 Main paradigms in health technology assessment.

Most value frameworks can be grouped together into three different *HTA paradigms*[10] (Fig. 21.2):

- The first paradigm, *economic evaluation*, is focused on the costs and benefits of a new technology and is centered around the notion of cost-effectiveness. Budget impact analysis (BIA) is also widely used alongside cost-effectiveness analysis in economic evaluation systems.
- The second paradigm, probably best termed as *comparative assessment* (although often referred to as *comparative effectiveness research* in the US context), is concerned with the added clinical benefit a new medical technology provides over its appropriately chosen comparator.
- The third paradigm encompasses various *multidimensional techniques*, e.g., scoring models and multicriteria decision analysis (MCDA). These are aimed at determining various clinical, economic, societal, and other sources of value for a new technology as well as their relative importance. Based on this, the main objective is prioritization for pricing and funding decisions.

Economic Evaluation

Economic evaluation has been the most widely used paradigm in countries in which the mainstream pharmacoeconomic tradition has developed or which have embraced it. Cost-effectiveness and budget impact are the core concepts in value assessment. *Cost-effectiveness* denotes a positive balance between the health benefits attainable through the new technology and the costs related to its application. If a new technology is cost-effective, it is "value-for-money" so it is worth for payers to consider buying it. The BIA identifies the costs and possible savings related to the application of a new technology and investigates the affordability of the new technology in function of available resources. Here the main question is if payers have enough money to purchase or fund the new technology. Cost-effectiveness and BIAs are interrelated; if a new technology is worth buying, it is still not sure that payers will have the necessary resources to buy it. Vice versa, if funds are available for a new technology, it may still not be worthwhile for investment.

Both cost-effectiveness analysis and BIA are quantitative techniques whose results may be influenced by the methodological choices that designers and users of HTA systems take. The cost-effectiveness of a new technology is usually determined by establishing its *incremental cost-effectiveness ratio* (defined as the incremental cost of providing one quality-adjusted life year [QALY] with the new technology as opposed to the cost of providing one QALY with an already available comparator technology) and by comparing it against a threshold value. Threshold values are often expressed as the multiple of the per capita gross domestic product[11] (e.g., in the United Kingdom or in Poland) or sometimes as the multiple of the minimum wage in a country (e.g., Slovakia). Although many threshold values are asserted as being based on prior value choices by payers, they are ultimately a matter of policy decision and their choice influences the outcomes of cost-effectiveness analyses. For example, "traditional" thresholds such as "X times per capita GDP" are mentioned as access barriers for distinct groups of health technologies such as orphan medicines, targeted combination therapies in oncology—hematology, or some special medical devices (e.g., implants). Because of this, the overwhelming majority of countries applying cost-effectiveness analysis do not use rigid thresholds. Rather, they have opted for differentiated thresholds or they incorporate cost/QALY results into a broader perspective or into a deliberative appraisal process.

The main methodological issue with BIA is the proper identification of relevant costs. We usually distinguish between *direct costs* which are directly related to the use of the new medical technology (e.g., drug administration cost, hospitalization cost) and *indirect costs* which may be medical and nonmedical and are more loosely linked to the technology (e.g., loss of productivity, dietary requirements, rehabilitation need, sick leave payments). Most of the BIAs are in fact just a financial budget sheet that compares cost differences between situations of including or not including a drug. The BIA shall also cover avoided direct and indirect costs, i.e., savings. For example, expenditures related to other medical treatments substituted by the new technology are direct savings, whereas "decreased loss" of productivity (i.e., increase in productivity) is indirect savings.[12]

Perspective is also a key issue in economic evaluation. Relevant costs and benefits are always defined and collected from the perspective of a specific healthcare stakeholder. For example, from a technical payer's perspective, indirect costs incurred by caretaker families are unlikely to be relevant unless the payer compensates the families for their efforts. However, from a policy-maker or societal perspective, these costs become relevant. Most healthcare systems require that cost-effectiveness analyses be performed from a payer perspective but some countries (most prominently Sweden and the Netherlands) also require the presentation of the societal perspective. In the case of BIA, countries usually insist on the payer perspective, and payers do not pay attention to indirect savings in the healthcare system.[13]

Economic evaluation strongly relies on *modeling,* especially when the data are limited. The main purpose of modeling is to extrapolate available clinical and economic evidence (e.g., clinical trial results as well as costs linked to the therapy) to the time frame which is relevant for decision-making. In addition, by incorporating metaanalyses and other sources of evidence, modeling makes it possible to carry out comparisons with comparators which were not originally included in clinical trials. For modeling purposes, the clinical problem or disease is usually "simplified" into a well-defined sequence of health states. For this simplification, decision-tree models, Markov models, and Monte Carlo simulations are the most frequently used techniques. Decision-tree models define mutually exclusive patient pathways between which patients cannot switch. Markov models are based on mutually exclusive health states with transition probabilities between these with time. Finally, Monte Carlo simulations enable patient-level analyses where individual factors of individual patients are incorporated into the analysis.[14] If the disease or condition evolves with time, Markov models or Monte Carlo simulations are preferred. In Markov models, time is represented by discrete periods called "cycles" and patients can move between health states with a certain transition probability, which depends only on the state they are in. Consequently, Markov models have no memory in contrast to Monte Carlo simulations, which can follow any changes in a patient pathway. Here, model inputs are drawn from statistical distributions.[15,16]

Economic evaluation has a highly developed methodological apparatus and its suitability for the comparison of very different technologies are strengths to be pointed out. Possible weaknesses can be its financial focus, resource-intensiveness, and the fact that for the interpretation of its results, policy-makers and payers are supposed to be acquainted with its methodology, which cannot always be the case in smaller or developing healthcare systems. Economic evaluation can exist in a "heavy" form in which a public agency performs both single technology assessments (STA) and comprehensive multiple technology assessments and in a "lighter" form where assessments are triggered by submissions by pharmaceutical companies (STA only).

The reference institution for the economic evaluation approach has traditionally been England's National Institute of Care and Health Excellence. Scotland uses the same approach in a different practical implementation by the Scottish Medicines Consortium. In the United Kingdom, because of the decentralized structure of healthcare budgets, economic evaluation has traditionally been focused on cost-effectiveness analysis rather than BIA but recently the latter has also gained importance. Other countries applying economic evaluation in more or less comprehensive forms include Poland, Hungary, South Korea, Thailand, and Brazil. Some developed healthcare markets such as Australia, Canada, Sweden, and the Netherlands use economic evaluation extensively as one component in a multidimensional decision-making framework. France, which is widely regarded as the "home country" for comparative benefit assessment (see below), introduced cost-effectiveness analysis for special cases (new medicines with high budget impact and claiming high added therapeutic benefit) in 2012.[17] A reverse example is found in South Korea, which generally uses cost-effectiveness analysis but does not require it in special cases if the new technology offers considerable price advantage. For the time being, many developing healthcare systems across Asia, the Middle East, and Latin America only require BIAs of new technologies. This is probably the focus of the assessment that has been on how to contain an already limited budget.

Comparative Assessment

Comparative assessment, sometimes called qualitative assessment or comparative effectiveness research (especially with reference to the United States[18,19]), bears more resemblance than economic evaluation in its approach to the evaluation

France (ASMR framework)	Germany (early benefit assessment system)
Exceptional added clinical benefit (ASMR I)	Major added benefit
Significant added clinical benefit (ASMR II)	Medium added benefit
Moderate added clinical benefit (ASMR III)	Moderate added benefit
Low added clinical benefit (ASMR IV)	Unquantifiable added benefit
No added clinical benefit (ASMR V)	No added benefit

FIGURE 21.3 Value category denominations in France and Germany.

logic of regulatory authorities. In this paradigm, sources of added clinical benefit are in the focus of HTA. The new technology, most commonly a new medicine, is compared in a structured, rigorous way to an already available comparator with special regard to clinical benefit. The comparator is usually the most widely used (gold standard) technology in any given country—for medicines, it can be another medicine, but if no other medicine is available, then it can be any other health technology (surgical procedure, radiotherapy, etc.) or even palliative care. Added clinical benefit should normally come from higher efficacy, but more favorable side effect (adverse event) profiles, convenience of use for patients (e.g., easier or less frequent administration), impact on patient adherence, and evidence about real-world effectiveness may also be considered. Countries using comparative assessment differ in the type of evidence they require to accept new technologies.

Systems of comparative assessment usually apply standard categories of added clinical benefit. Based on the assessment performed, they may categorize new technologies as having "exceptional," "significant," "minor," or "no" added clinical benefit. Exact denominations and definitions of such "value categories" differ from system to system (Fig. 21.3). Formal scoring systems are uncommon. Rather, the value category of a medicinal product is established through collective expert judgment. Some comparative benefit assessment systems have established formal scientific committees for value category designation (e.g., Transparency Commission in France, Technical-Scientific Commission in Italy). These committees usually consist of clinical scientists and pharmacologists and use deliberative processes and techniques to reach their conclusions. These can be recommendations toward policy-makers and payers as well as formal decisions.

The economic viewpoint is included typically indirectly in comparative assessment systems; the added clinical benefit of a new technology (as expressed in the value category) will determine the eligibility of the new technology for public funding ("reimbursability") as well as its maximum allowable price. For example, for new medicines which are granted a designation of exceptional added clinical benefit in France, the marketing authorization owner can claim a reasonable price premium over already funded medicines. On the other hand, medicines with minor or no added clinical benefit must be priced at par with, or even at discount to, already available medicines. Budget impact considerations are usually not part of HTA in such systems—rather they are a part of pricing and reimbursement negotiations between payers and technology owners.

Comparative assessment has the potential to deliver appraisals and recommendations for being more intelligible and relevant for policy-makers and payers than economic evaluation reports. Clinical benefit is explicitly assessed; assessment criteria and the deliberative assessment method can be customized to local health system characteristics. As opposed to these advantages, possible shortcomings can be insufficient standardization and transparency in the application of value assessment criteria and the exclusion of cost-effectiveness aspects of HTA.

Comparative assessment is applied in some major European pharmaceutical markets such as France, Italy, and Germany. Japan, too, has traditionally used an assessment logic similar to the comparative assessment but is currently underway to incorporate economic considerations more emphatically, as France included of cost-effectiveness analysis for selected products.

Multidimensional Value Assessment Approaches

Multidimensional value assessment approaches do not constitute a homogeneous category of HTA methods. Rather, they comprise a set of different techniques which have in common that they acknowledge and assess different dimensions of value for new technologies. Typically, economic and clinical sources of value are considered, but it is also common for multidimensional HTA systems to ponder societal and ethical considerations.

Regarding practical applications, Canada, Sweden, Australia, and the Netherlands all have formal HTA systems which show some elements of "multidimensionality," at least for medicines. The Canadian framework operated by the Canadian Agency for Drugs and Technologies in Health shows a high degree of integration of clinical and economic considerations.

1. Added clinical benefit	Criteria which relate to the performance of a new medicine in its requested indication* along main possible sources of clinical benefit (e.g. efficacy, safety, convenience, adherence, real-world effectiveness etc.)
2. International funding and assessment references	Criteria which relate to 1) relevant health technology appraisals in the requested indication in reference international pharmaceutical markets (both economic evaluation and comparative assessment) and 2) international reimbursement / public funding references.in the requested indication
3. Health policy alignment	Selected criteria which describe how much the new medicine in its requested indication supports the implementation of national (and eventually regional) health policy priorities
4. Social and ethical considerations	Selected social and ethical considerations the new medicine is able to address in the country
5. Budget impact	Net reimbursement outflow that the new medicine is expected to generate in its requested indication in various stages of its adoption (budget impact mitigation effects through managed entry agreements are taken into consideration)

* requested indication = indication in which the manufacturer requests reimbursement / public funding for the medicine

FIGURE 21.4 Example of value dimensions in a pragmatic value assessment system.[21]

In Sweden, societal and ethical considerations are formally incorporated into the HTA framework alongside cost-effectiveness and BIA. In Australia, economic evaluation is dominant, but the HTA system operated by the Pharmaceutical Benefits Advisory Committee allows for "rules of rescue" whereby non-cost-effective products can also be included in the national reimbursement formulary. In the Netherlands, therapeutic benefit, cost-effectiveness analysis, and BIA are all parts of a HTA system, and products must meet the combined requirements to be eligible for reimbursement.

MCDA has recently been adapted as a scientific approach to multidimensional value assessment.[20] MCDA is a set of techniques which apply several assessment criteria with different weights elicited from experts through collective brainstorming or decision-making techniques. Although the MCDA process itself can be reasonably standard, it can be resource-consuming and is better suited for individual decision problems than for continuous pricing and reimbursement decisions. As an alternative, pragmatic value assessment models such as balanced assessment systems[21] (Fig. 21.4) propose standard value dimensions (e.g., added clinical benefit, health policy alignment, social and ethical considerations, affordability, etc.) with customizable sets of decision-making criteria and weights (scores) to facilitate funding and purchasing decisions. These often focus on healthcare systems where there are few capabilities or resources for the deployment of extensive HTA systems. Related to this, resource-intensive quantitative analyses may be substituted through referencing to previous international assessments with the consideration of adoption feasibility. Such pragmatic approaches are best suited to help ranking and prioritization decisions based on multiple policy-relevant criteria. Among practical HTA applications, Taiwan is sometimes mentioned as a resource-conscious, "affordable" system in which comparative effectiveness, budget impact, and cost-effectiveness considerations are analyzed.[22]

OBJECTIVE 21.3. DESCRIBE A HIGH-LEVEL HEALTH TECHNOLOGY ASSESSMENT PROCESS

In any country using HTA, assessment follows a process which can be very precisely defined (like in the United Kingdom, Germany, or Sweden) or less precisely regulated (as in several upper middle-income countries across Central Europe or Latin America using elements of HTA). Fig. 21.5 shows an example for a structured HTA process (embedded into the wider pricing and reimbursement process) for new technologies seeking public funding. It commences with the submission of a dossier by the technology owner followed by technical assessment, then possible consultations (public hearings) with stakeholders, then formal appraisal, then pricing and reimbursement negotiations, and the final decision. It must be noted that such a process can be established in very different organizational setups (i.e., different allocation competences between participating stakeholders and responsible bodies).

A key principle in the HTA process is the separation of the assessment and appraisal.[23] *Assessment* is a technical activity carried out by specialized experts that comprises of the review of the technology owner's submission and the available evidence. The output of the assessment phase is usually an HTA report, which is sent to those participating in the *appraisal* phase. Appraisal is the analysis and consideration of the results of technical assessment, eventually together with other decision-relevant aspects or circumstances which are not covered by the assessment phase. The appraisal phase is often executed by a specialized committee and may conclude with a formal recommendation to the pricing and

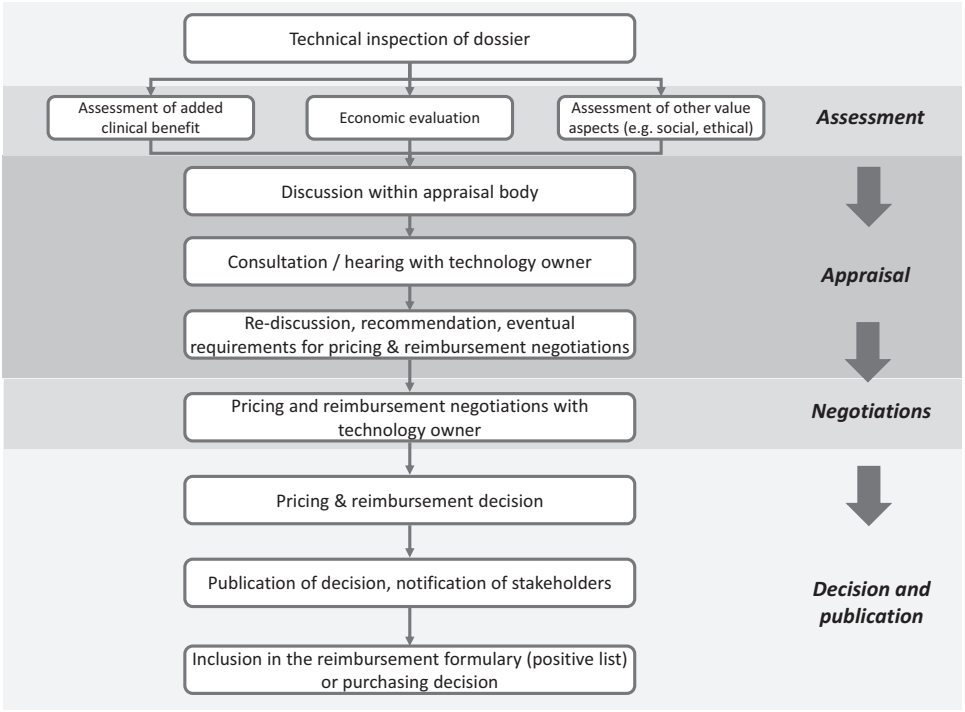


FIGURE 21.5 Example of health technology assessment process for new health technologies seeking public funding in a system applying multidimensional assessment and negotiations with technology owners.

reimbursement decision-making body which then conducts pricing and reimbursement negotiations with the technology owner. Less typically, appraisals can be linked to a formal binding decision which is not followed by any further negotiations. Assessment, appraisal, recommendation, and decision competences are organized in different forms across various healthcare systems.

As mentioned above, the HTA process flows directly into the pricing and reimbursement negotiation process in several healthcare systems. In systems using comparative benefit assessment, appraisal usually takes the form of a value category designation and subsequent pricing and reimbursement negotiations are determined and guided by this. In many countries (e.g., Belgium, Italy, Poland, Hungary), pricing negotiations are geared toward *managed entry agreements* (MEAs) which are contracts between payers and technology owners which contain cost-sharing or risk-sharing elements. Through MEAs payers can attain net prices which are lower than list prices in a confidential way, whereas technology owners can protect their products against international reference pricing mechanisms and parallel exports from systems with low prices to systems with higher prices.

OBJECTIVE 21.4. EXPLAIN HEALTH TECHNOLOGY ASSESSMENT SYSTEM DESIGN IN CURRENT PRACTICE

Although the academic literature of HTA is multidisciplinary and multifaceted, the choice and design of HTA systems has traditionally been a technocratic activity in which HTA systems are designed based on HTA guidelines and best practice recommendations. In this design approach, international best practice is structured, generalized, and codified in a way which gives guidance to designers of HTA systems on what newly implemented systems should look like.[24] This traditional design approach, however, is not without possible risks or drawbacks. As it does not adequately consider the specific local institutional requirements of individual healthcare systems, cultural and political fit may be limited. Furthermore, as best practice recommendations are overwhelmingly based on practices of developed healthcare systems, they may result in resource-intensive implementations in smaller or emerging systems which will not be sustainable for the lack of expertise, available healthcare data, or sufficient funding.

These risks may push a different, institutional design approach into the forefront. From this "institutional" approach, HTA systems can be designed in a rational and structured way which proceeds from context assessment though model design to process design. A high-level overview of this process is shown in Fig. 21.6.

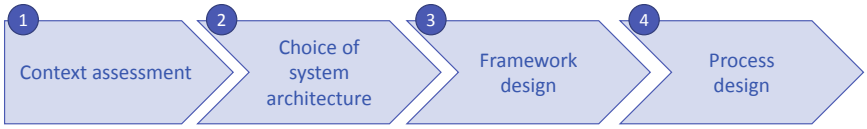


FIGURE 21.6 Health technology assessment design process from an institutional approach.

1. The process starts with context assessment. In this phase, variables of the institutional environment are identified and described. Some of the most commonly analyzed attributes are as follows:
   a. Policy objectives to be fulfilled by the introduction of HTA
   b. Availability of human resources for assessment and appraisal
   c. Level of health economics expertise available (e.g., preexisting knowledge, knowledge of international HTA models)
   d. Availability of health data (e.g., scope and quality of data available for HTA)
   e. Availability of funding for HTA (e.g., amount of annual funding, sustainability of funding)
   f. HTA governance (e.g., existing legal frameworks, level of centralization, political and policy requirements, decision-making styles)
   g. Institutional robustness (e.g., intensity of political cycles, influence from different stakeholders)

   Ideally, the first phase concludes with a comprehensive and dynamic description of the institutional environment in the form of a status report.
2. The second (system architecture) phase addresses high-level conceptual design questions. It includes the choice of the HTA paradigm (economic evaluation, comparative benefit assessment, some multidimensional HTA approach), the perspective of assessment, the timing of assessment (pricing and listing decisions only or also price and reimbursement reviews), principles of HTA governance, technology scope (e.g., pharmaceuticals or broader scope), and eventual data transferability principles.
3. The third (framework design) phase covers a range of delicate technical design decisions. Assessment criteria must be selected, definitions and metrics must be specified, and eventually scores and the weights need to be defined (in systems using multiple criteria). These decision rules will govern if a new technology can be admitted into public funding without conditions or only with certain conditions or it will not be admitted into public funding at all.
4. In the fourth phase, the HTA process is to be designed properly. Milestones and responsibilities as well as the tasks of participating stakeholders must be defined. Internal and external assessment reports must be developed and publicly available information about the HTA process and its outputs (reports, recommendations) must be delineated. In addition, IT systems supporting HTA assessment must be outlined.

CONCLUSION

HTA has become a core system to assist healthcare decision-making under resource constraints in many countries around the world. As mentioned above, a lot of systems have been or are currently deployed in middle-income and emerging healthcare systems, often reflecting on the practices in the United Kingdom, Canada, and Australia. Regardless of the exact architecture that an HTA system takes in any given healthcare system, it is essential that HTA implementation itself should be regarded as a health policy intervention. As such its benefits for policy-makers, payers, and the wider society should exceed the direct and indirect costs that are incurred through its design and implementation. This means that HTA should be cost-effective itself. From a payer perspective, cost-effectiveness of HTA means that the payer benefit (higher quality of decisions) from the increased quality of information should exceed the cost of producing that information. From a societal perspective, health outcomes from improved system efficiency (and possibly higher transparency) through HTA should exceed the direct and indirect costs of HTA. This "*HTA cost-effectiveness requirement*" can be implemented through various system architectures with various scopes and focuses. It is imperative, however, that the chosen architecture should be in line with the determining elements of the local institutional environment.